OVER A LOCALLY NOETHERIAN BASE AN IMPURITY CAN BE REPLACED BY ONE WITH A QUASI-FINITE MORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Replacement of an impurity by one with a locally finitely presented morphism and a closed point. Same chapter flat.tex, label lemma-impure-finite-presentation, invoked at proof line 4709. Provides the starting impurity with the morphism locally of finite presentation and the specialisation point closed in the fibre.
- Spreading an associated point to a neighbourhood, and the existence of a quasi-finite closed subscheme through a nearby point. More on Morphisms, labels more-morphisms-lemma-relative-assassin-in-neighbourhood (proof line 4717), more-morphisms-lemma-nonempty-generic-fibre (proof line 4721) and more-morphisms-lemma-quasi-finite-quasi-section-meeting-nearby-open-X (proof line 4724). The first produces an open set on which generic points of fibres are associated points of the relative situation, the second gives an open subset of the base contained in the image of that open set, and the third cuts out a closed subscheme through the specialisation point which is quasi-finite there and meets the chosen open set away from the fibre over the base point.

Source reference. Original lemma 05FX: sources/stacks-flat/pinned-original-source.tex, source lines 4514–4514. Navigation pointer for \ref{lemma-impure-finite-presentation} used inside the counted span (line 4709); the argument that consumes it is retained in the span.

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Source reference. Original lemma 05F5: sources/stacks-chow/tags.txt, source lines 7014–7014. Navigation pointer for \ref{more-morphisms-lemma-nonempty-generic-fibre} used inside the counted span (line 4721); the argument that consumes it is retained in the span.

Source reference. Original lemma 05GU: sources/stacks-chow/tags.txt, source lines 7073–7073. Navigation pointer for \ref{more-morphisms-lemma-quasi-finite-quasi-section-meeting-nearby-open-X} used inside the counted span (line 4724); the argument that consumes it is retained in the span.

Source reference. Original lemma 05F1: sources/stacks-chow/tags.txt, source lines 7010–7010. Navigation pointer for \ref{more-morphisms-lemma-relative-assassin-in-neighbourhood} used inside the counted span (line 4717); the argument that consumes it is retained in the span.

Source reference. Original reference 05FW: sources/stacks-flat/pinned-original-source.tex, source lines 4473–4473. Navigation pointer for \ref{situation-pre-pure} used inside the counted span (line 4700); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05FW. Assume that S is locally Noetherian. If there exists an impurity of $\mathcal{F}$ above s, then there exists an impurity $(g : T \to S, t' \rightsquigarrow t, \xi)$ of $\mathcal{F}$ above s such that g is quasi-finite at t.*

Proof. We may replace S by an affine neighbourhood of s. By Lemma 05FX we may assume that we have an impurity $(g : T \to S, t' \rightsquigarrow t, \xi)$ of such that g is locally of finite type and t a closed point of the fibre of g above s. We may replace T by the reduced induced scheme structure on $\overline{\{t'\}}$. Let $Z = \overline{\{\xi\}} \subset X_T$. By assumption $Z_t = \emptyset$ and the image of $Z \to T$ contains t'. By More on Morphisms, Lemma 05F1 there exists a nonempty open $V \subset Z$ such that for any $w \in f(V)$ any generic point ξ' of V_w is in $\mathrm{Ass}_{X_T/T}(\mathcal{F}_T)$. By More on Morphisms, Lemma 05F5 there exists a nonempty open $W \subset T$ with $W \subset f(V)$. By More on Morphisms, Lemma 05GU there exists a closed subscheme $T' \subset T$ such that $t \in T'$, $T' \to S$ is quasi-finite at t, and there exists a point $z \in T' \cap W$, $z \rightsquigarrow t$ which does not map to s. Choose any generic point ξ' of the nonempty scheme V_z. Then $(T' \to S, z \rightsquigarrow t, \xi')$ is the desired impurity. $\square$